Code No: 156DY **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, February - 2023
QUANTITATIVE ANALYSIS FOR BUSINESS DECISIONS
(Common to CE, EEE, CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a) What are the advantages and limitations of Linear Programming models? [2]
b) Explain what is meant by "shadow price of the resource" in the context of duality. [3]
c) What is assignment Problem? Give two areas of its applications. [2]
d) Give the mathematical formulation of transportation problem. [3]
e) How would you deal with replacement of items that fail completely and suddenly? [2]
f) What is the difference between group replacement and individual replacement? Under which circumstances group replacement may become profitable? [3]
g) How is the concept of dominance used in simplifying the solution of rectangular games? [2]
h) Explain the difference between expected opportunity loss and expected value of perfect information. [3]
i) What happens to the Queue when arrival frequency is more than servicing frequency? [2]
j) What are the advantages and disadvantages of simulation models? [3]

PART – B

(50 Marks)

2.a) Discuss the assumptions in Linear Programming Problem.
b) Min $z = x_1-3x_2+2x_3$, subject to
$3x_1-x_2+3x_3 \leq 7$
$-2x_1+4x_2 \leq 12$
$-4x_1+3x_2+8x_3 \leq 10$ and
$x_1, x_2, x_3 \geq 0$. [10]

OR

3. Maximize $z=4x_1+5x_2-3x_3+50$, subject to the constraints,
$x_1+x_2+x_3=10$
$x_1-x_2 \geq 1$
$2x_1+3x_2+x_3 \leq 40$
$x_1, x_2, x_3 \geq 0$. [10]

4. Goods have to be transported from factories F_1, F_2, F_3 to ware-houses W_1, W_2, W_3 and W_4. The transportation cost per unit, capacities and requirements of the warehouse are given in the following table.

	W_1	W_2	W_3	W_4	Capacity
F_1	95	105	80	15	12
F_2	115	180	40	30	7
F_3	195	180	95	70	5
Requirement	5	4	4	11	

Find the distribution with minimum cost. [10]

OR

5. Five men are available to do five different jobs. From past records, the time (in hours) that each man takes to do each job is known and given in the following table:

	Job				
Man	**I**	**II**	**III**	**IV**	**V**
A	2	9	2	7	1
B	6	8	7	6	1
C	4	6	5	3	1
D	4	2	7	3	1
E	5	3	9	5	1

Find the assignment of men to jobs that will minimize the total time taken. [10]

6. A firm has a machine whose purchase price is Rs.20, 000. Its maintenance cost and resale price at the end of different years is as given below:

Year	**1**	**2**	**3**	**4**	**5**	**6**
Maintenance cost(Rs.)	1500	1700	2000	2500	3500	5500
Resale price(Rs.)	17000	15300	14000	12000	8000	3000

a) Obtain the economic life of machine and the minimum average cost.

b) The firm has obtained a contract to supply the goods produced by the machine, for a period of 5 years from now. After this time period, the firm does not intend to use the machine. If the firm has a machine of this type that is one year old, what replacement policy should it adopt if it intends to replace the machine more than once? [10]

OR

7. A large TV manufacturer has to provide "Service after sales" on annual contract basis. The company has now 500 TV sets; and has to decide on service policy. Service is restricted to replacement of minor parts. The failure pattern of these minor parts is a sunder:

Month	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Number failing after month of service	10	20	60	130	160	180	40

It costs Rs.50 to replace minor parts as and when they fail. It costs Rs.20 to replace minor parts on a planned service basis, when parts are replaced (by fixed interval visits) even before failure. Which policy would you recommend?

a) Replace as and when minor parts fail.

b) Replace at periodic intervals, say at the end of one month/two months etc? [10]

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8. Solve the following game using graphical approach: [10]

		B's Strategy			
		b_1	b_2	b_3	b_4
A's strategy	a_1	8	5	-7	9
	a_2	-6	6	4	-2

OR

9. A TV dealer finds that the cost of holding a TV in stock for a week is Rs.50. Customers who cannot obtain new TVs immediately tend to go to other dealers and he estimates that for every customer who cannot get immediate delivery he loses an average of Rs.200.For one particular model of TV the probabilities of selling 0, 1,2,3,4 and 5 TVs in a week are 0.05,0.10,0.20,0.30,0.20 and 0.15 respectively.
a) How many televisions per week should the dealer order? Assume that there is no time lag between ordering and delivery.
b) Compute EVPI.
c) The dealer is thinking of spending on a small market survey to obtain additional information regarding the demand levels, how much should he be willing to spend on such a survey? [10]

10. A commercial bank has two tellers working on savings accounts. The first teller handles withdrawals only. The second teller handles deposits only. It has been found that service time distributions for both deposits and withdrawals are exponential with mean service time are 3 minutes per customer. Depositors are found to arrive in a Poisson fashion throughout the day with mean arrival rate of 16 per hour. Withdrawers also arrive in a Poisson fashion with mean arrival rate of 14 per hour. What would be the effect on the average waiting time for depositors and withdrawers if each teller handles both withdrawals and deposits? [10]

OR

11. A company manufactures around 150 two wheelers. The daily production varies from 146 to 154 depending upon the variability of raw materials and other working conditions.

Production per day	**Probability**
146	0.04
147	0.09
148	0.12
149	0.14
150	0.11
151	0.10
152	0.20
153	0.12
154	0.08

The finished two wheelers are transported in a specially arranged vehicle accommodating only 150 two wheelers. Using the following random numbers 80,81,76,75,64,43,18,26,10,12,65,68,69,61,57 simulate the process to find on:
a) What will be the average number of two wheelers waiting in the factory?
b) What will be the average number of empty spaces on the specially designed vehicle? [10]

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